

Full Length Article

Safeguarding IoT consumer devices: Deep learning with TinyML driven real-time anomaly detection for predictive maintenance

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ABSTRACT

Internet of Things (IoT) security is paramount for enterprises, as it includes several strategies, techniques, actions, and protocols that aim to alleviate the high vulnerability of cutting-edge businesses. IoT consumer devices, from smart home appliances to wearable gadgets, have become ubiquitous daily, facilitating automation and seamless connectivity. However, ensuring their reliability and security presents a tremendous challenge. Anomaly detection methods offer a promising solution, especially those powered by TinyML (Machine Learning (ML) on Tiny Devices). These IoT devices can autonomously identify unusual behaviours or patterns that diverge from regular operation by leveraging the proficiencies of deep learning (DL) techniques enhanced for resource-constraint environments, like neural networks. Incorporating DL, anomaly detection, and TinyML allows real-time monitoring and proactive mitigation of malfunctions or security breaches in IoT devices. This advanced technology ensures improved reliability, privacy, and overall user experience in the dynamic landscape of connected devices, whether identifying irregular health data or detecting unauthorized access attempts on a smart door lock from the wearable fitness tracker. Therefore, this study develops a new Deep Learning technique to secure IoT consumer devices with TinyML Driven Real-time Anomaly Detection for Predictive Maintenance (DLTML-RTADPM). The DLTML-RTADPM technique aims to recognize and categorize the anomalies in IoT consumer devices. At the primary phase, the DLTML-RTADPM model normalizes the input data using Z-score normalization. In the DLTML-RTADPM method, the Fennec Fox Optimization Algorithm (FFA) is used for a high dimensionality reduction process where the optimal feature set is chosen. The DLTML-RTADPM technique implements gradient least mean squares with a bidirectional long short-term memory (GLMS-BiLSTM) approach for anomaly detection. To further improve the detection results of the DLTML-RTADPM technique, the Jaya optimization algorithm (JOA)-based hyperparameter tuning process is utilized. A series of simulations are performed on the benchmark dataset to ensure better detection outcomes of the DLTML-RTADPM model. The investigational validation of the DLTML-RTADPM method portrayed a superior accuracy value of 98.11% over other techniques.

1. Introduction

In the digital era, the attack surface for user networks was growing rapidly with the tremendous growth of IoT devices. Companies that imitated utilizing anomaly recognition software and tools earlier the pandemic struck were highly organized to recognize the speedy and ever-fluctuating desires of their users [1]. They could better predict the upsurge in e-commerce transactions and the growth in online shopping, making it very easy for them to affiliate with the main concern and transport their consumers to an excellent online shopping experience [2]. A malicious attack can be executed to influence the computing

power of an IoT system, or it can be executed to damage a method by creating false data. For the first, this could be evident as additional inaccurate network traffic; for the second, this can be noticeable as false messages being approved as legitimate [3]. Furthermore, anomalies can occur instantly or in a few windows of time. A fast temperature change in a sensor of temperature from one step to another might specify a strict sensor fault or an environmental change, but a slow change over the sequence of hours or days might be the usual operating behaviour. As any motor runs, its temperature slowly alters [4]. This might be nominal behaviour, but a fast upsurge in temperature or a temperature beyond some threshold would be anomalous. Exact methods are measured,

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concentrating on anomaly recognition in time-series data [5].

Mainly, ML and DL are standard tools for anomaly recognition. In present years, IoT devices are one of the most excellent and most proficient in implementing compound computation tasks [6]. This development has allowed us to run innovative techniques on the network's edge. Directing ML and data analysis processes at the edge of the network aids in averting stays while moving data from the basis to the cloud, leading to quick computational time [7]. Uniting AI abilities with IoT devices was directed to the Artificial Intelligence of Things (AIoT) model. In the IoT framework, ML-linked tasks were usually executed on the cloud outcome with higher potential and less than desired Quality of Service (QoS). IoT devices gather information and direct it to the cloud server. It restricts the real analytic skills of the method [8]. Moreover, there are bandwidth, safety, and privacy tasks. TinyML refers to these problems by utilizing ML methods on lower-power devices like the MicroController Unit (MCU) by processing resources and restricted memory [9]. In the present scenario, TinyML has developed as a novel tendency that combines ML methods into lesser, lower-power plans motorized by MCU. The importance of TinyML stays in its capability to utilize influential methods in battery-driven devices, thus permitting real-time data processing and increasing receptiveness while decreasing the energy utilization needed for wireless communication [10]. This technology was predicted to experience exponential development in future years and obtain extensive acceptance.

This study develops a new Deep Learning with TinyML Driven Real-time Anomaly Detection for Predictive Maintenance (DLTML-RTADPM) technique to secure IoT consumer devices. The DLTML-RTADPM technique aims to recognize and categorize the anomalies in IoT consumer devices. At the primary phase, the DLTML-RTADPM model normalizes the input data using Z-score normalization. In the DLTML-RTADPM method, the Fennec Fox Optimization Algorithm (FFA) is used for a high dimensionality reduction process where the optimal feature set is chosen. The DLTML-RTADPM technique implements gradient least mean squares with a bidirectional long short-term memory (GLMS-BiLSTM) approach for anomaly detection. To further improve the detection results of the DLTML-RTADPM technique, the Jaya optimization algorithm (JOA)-based hyperparameter tuning process is utilized. A series of simulations are performed on the benchmark dataset to ensure better detection outcomes of the DLTML-RTADPM model. The key contribution of the DLTML-RTADPM model is listed below.

- The DLTML-RTADPM model utilizes Z-score normalization to preprocess the data, transforming it into a standard distribution with a mean of 0 and a standard deviation of 1. This step ensures all features are on the same scale, preventing bias due to differing feature magnitudes. Standardizing the input data improves the method's ability to learn from all features effectively.
- The FFA technique is utilized to optimize feature selection, inspired by the fennec fox's survival strategies. This approach effectively detects the most relevant features, improving the method's capability to concentrate on key information. By choosing optimal features, the model enhances efficiency and accuracy in anomaly detection.
- The GLMS-BiLSTM model integrates the gradient least mean squares algorithm with a BiLSTM network for effective anomaly detection. This hybrid methodology captures both temporal dependencies and bidirectional context, improving the model's accuracy in identifying anomalies. By incorporating these methods, the model improves its performance detecting complex patterns.
- The JOA is implemented to fine-tune the hyperparameters of the GLMS-BiLSTM model, optimizing its performance. This nature-inspired technique effectively searches for the optimum parameter values, improving the model's anomaly detection capabilities. Tuning the hyperparameters improves the model's accuracy and robustness in handling complex data.

- The DLTML-RTADPM approach integrates Z-score normalization, the FFA model for feature selection, GLMS-BiLSTM for anomaly detection, and JOA for hyperparameter tuning, creating a cohesive framework. This unique incorporation utilizes the merits of every technique, improving the model's ability to detect anomalies. The novelty is in the seamless integration of these advanced methods, each contributing to enhanced performance and making the model more efficient and accurate in anomaly detection tasks.

The article is structured as follows: Section 2 presents the literature review, Section 3 outlines the proposed method, Section 4 details the results evaluation, and Section 5 concludes the study.

2. Literature review

Karras et al. [11] present traditional TinyML techniques intended and proposed to enhance Big Data management in huge-scale IoT methods. This technique called EdgeClusterML, TinyCleanEDF, CacheEdgeML, TinyHybridSenseQ, and CompressEdgeML, which function organized to improve data process, quality control, and storage in IoT systems, using the abilities of Edge AI. TinyCleanEDF mainly uses federated learning (FL). EdgeClusterML unites reinforcement learning (RL) with self-organized maps. CompressEdgeML utilizes neural networks for adaptive data density. In [12], an ML-based anomaly recognition method that utilizes a retrofitting kit dependent upon a cost-effective and constrained IoT device is projected. Despite its restricted resources, the latter implements an advanced unsupervised anomaly recognition model locally, separately acquiring the normality performance of the observed asset. Moreover, to enhance the transparency of the observing procedure, the method suggests influencing BC technology as a non-refusable data source. Antonini et al. [13] present and estimate an endwise flexible and configurable anomaly recognition method that utilizes the IoT, Tiny-MLOps, and edge computing methods in a greatly developed atmosphere. The method turns on an IoT detecting Kit, dependent upon MicroPython firmware and ESP32 microcontroller, near the data source. The anomaly recognition technique utilizes the isolation forest system. Moreover, the method uses BC technology to deliver a clear and specific source of anomalies. In [14], a new structure, DroneSSL, is developed, combining the spatial crowdsourcing model with TinyML to improve anomaly recognition in the Internet of Drone Things (IoDT). This advanced model influences drones and unmanned ground vehicles (UGVs) for extensive data groups in atmospheres that are normally unreachable or dangerous. By using lightweight ML techniques besides innovative communication technologies, DroneSSL exceeds classical spatial-temporal data analysis models. Barbieri et al. [15] concentrate on presenting small transformer-based AD devices to create possible solutions for on-device AD. Beginning with the advanced Anomaly Transformer (AT) method, a small AD structure is developed, and an enhanced outline of the AT technique is discovered. It is utilized to create a compacted form attuned to resource-constrained IoT methods. A knowledge condensation device was projected to get a remarkably compacted AT method without reducing the performance of AD. Banerjee and Ghose [16] projected an improved lightweight model for accurately identifying usual Atrial Fibrillation (AF), sinus rhythm, and other abnormal heart rhythms utilizing single-lead ECG. A deep ResNet framework with an attention module was projected as the base method, mainly compacted adequately using a set of collaborative optimizer models.

Ismail et al. [17] presented a hybrid safety structure for SC methods, BC-DRLzSC, that combines BC and DRL to function in a zero trust (ZT) atmosphere. Notably, this method suggests a decentralized BC-based technique combined with smart contracts to achieve system participant registration and verification to control access and system resources. BC-DRLzSC accepts a ZT structure to strengthen SC safety, and DRL is used to develop a practical attack recognition technique. Hayajneh et al. [18], an endwise solution is projected that uses TinyML for the lower-

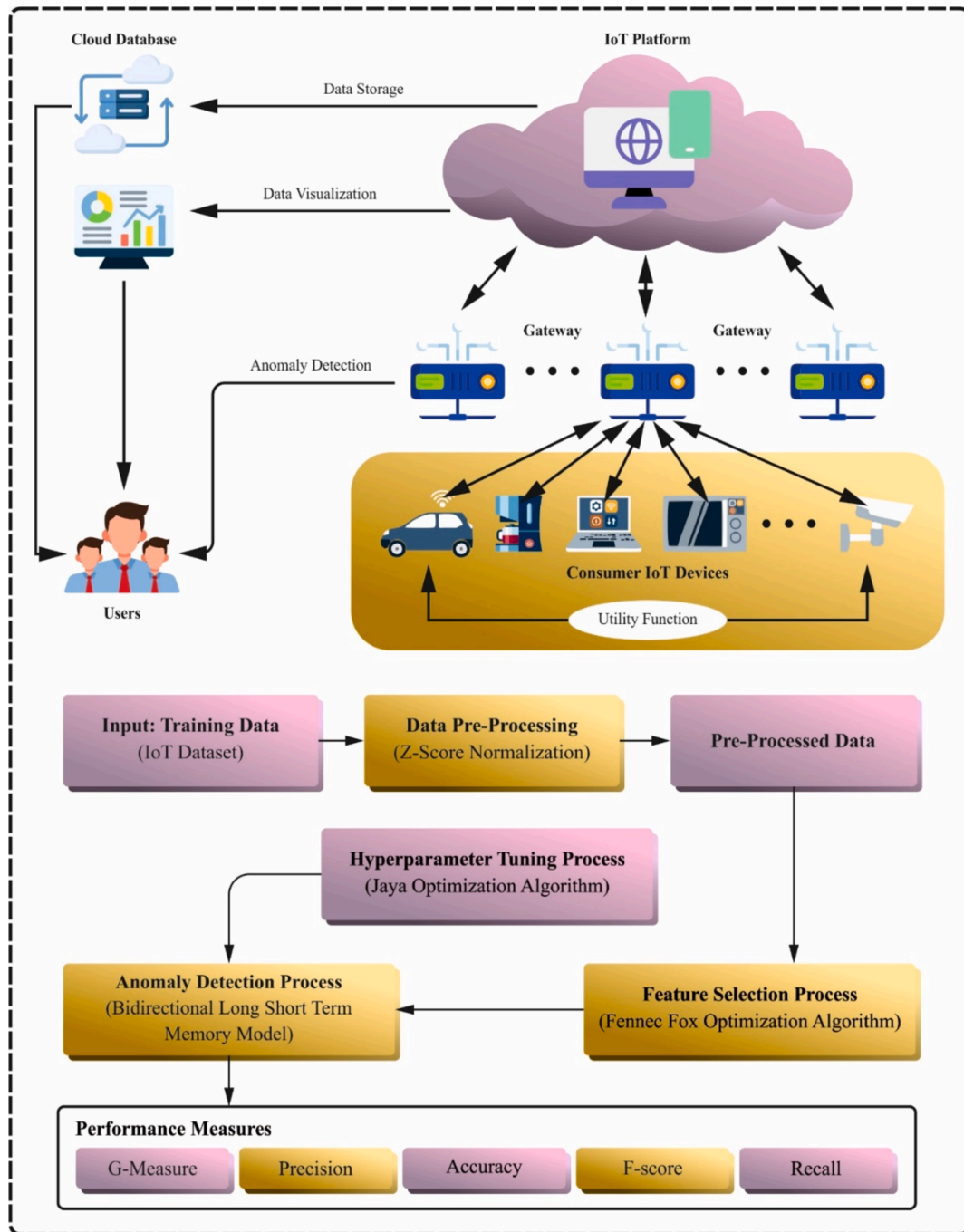


Fig. 1. Overall flow of DLTML-RTADPM technique.

cost acceptance of ML in the task of classification. This technique used image segmentation models employing morphological segmentation. Next, a CNN has been trained to categorize the features and class of fruits, thus improving the efficacy of the olive post-harvest procedure. In [19], the Adaptive Multipath Multichannel Energy Efficient (AMMEE) routing strategy is presented, which optimizes route selection based on energy consumption, bandwidth, queue length, and channel utilization. It uses multichannel and multipath techniques to mitigate collisions and balance network load, improving reliability. Alajlan and Ibrahim [20] apply TinyML to driver drowsiness detection, presenting an overview of TinyML and proposing five lightweight DL models for microcontroller deployment. Khatoun et al. [21] aim to enhance road anomaly detection using TinyML integrated with edge computing, demonstrating improved efficiency and accuracy through optimized Binary Neural Networks

(BNN) for remotely sensed images. Yuan et al. [22] introduce the ESP32-CAM, an affordable and user-friendly IoT platform integrated with a TinyML-based neural network model for detecting human emotions. The platform, optimized for IoT devices with limited computational power, provides students with hands-on experience in TinyML, enhancing their practical skills in the classroom. Tharayil et al. [23] integrate convolutional layers, LSTM cells, and attention layers in a novel neural network architecture, training the model on historical drilling equipment data with optimized deep-learning hyperparameters to address its unique characteristics.

Chiurato [24] explores using artificial intelligence, specifically TinyML and neural networks, for classifying accelerator pedal signals in electric vehicles. Le and Nguyen [25] present an innovative unsupervised learning approach using a low-cost microcontroller unit (MCU) to

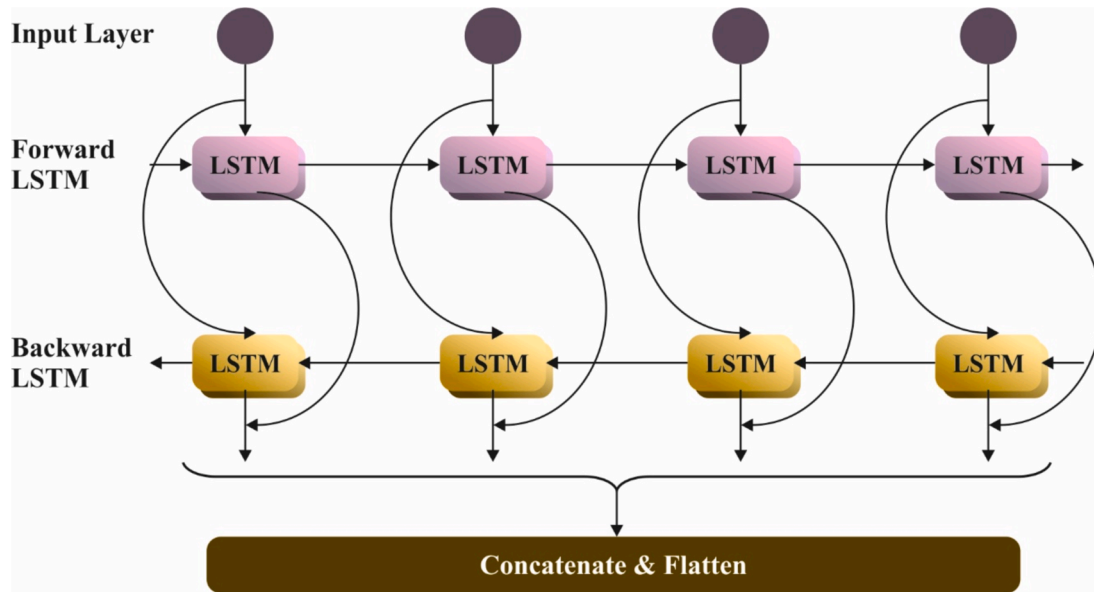


Fig. 2. Architecture of BiLSTM.

Table 1
Details of the dataset.

Type of Event	No. of Data Record
Backdoor	2500
DoS	2500
DDoS	2500
Injection	2500
MITM	1000
Scanning	2500
Ransomware	2500
Password	2500
XSS	2500
Normal	2500
Total Data Record	23,500

detect equipment abnormalities in fan vibration signals. This research develops a hybrid deep learning-based anomaly detection model that uses sequential data reconstruction to identify anomalous patterns without data labelling. Anidjar et al. [26] propose a real-time framework for anomaly detection in drones using sound analysis, incorporating a Transformer with a Convolutional Neural Network (CNN) model inspired by VGG architecture. Medeiros et al. [27] propose a multi-layered data processing methodology for edge computing, combining soft sensors, TEDA, and incremental clustering to detect driver behaviour. Jain et al. [28] introduce TinyVers, a low-power ML system-on-chip (SoC) designed for edge intelligence. It integrates a RISC-V processor, a 17 TOPS/W reconfigurable ML accelerator, a 1.7- μ W wake-up controller, and embedded eMRAM for efficient smart sensing and power management. Razak, Yogarayan, and Ullah [29] propose an IoT-based tool that monitors a driver's heart rate and breath for alcohol detection. The tool, integrated with a DC motor, powers down the motor and sends an SMS with the driver's location to an emergency contact if elevated heart rate and alcohol levels are detected. [30] presents a hybrid DL-based anomaly detection model that identifies anomalous patterns without requiring data labelling, enabling preemptive downtime prediction and enhancing smart factory efficiency through real-time service framework deployment. Srisang et al. [31] analyze the performance of a low-cost PM2.5 sensor under various outdoor conditions and assess the efficiency of seven calibration methods to improve its data accuracy by comparing it to a reference sensor. Asutkar et al. [32] present a transfer learning (TL) framework with a TinyML-powered CNN architecture for vibration-based fault diagnosis across diverse

machines. de la Fuente et al. [33] introduce the Edge Sensor Network for Predictive Maintenance (ESN-PdM) framework, a scalable, adaptive solution for real-time predictive maintenance of mining machinery, optimizing inference location based on accuracy, latency, and energy consumption using TinyML.

The limitations of the existing studies include the reliance on constrained hardware and resource-limited IoT devices, which affect the implementation of more complex ML models with higher accuracy and lower latency. Many of the proposed models, specifically those using DL or neural networks, face difficulty with the tradeoffs between model complexity and performance, as seen with the problems in achieving high accuracy on low-power devices. Additionally, inadequate quantization techniques for sophisticated models like LSTM or GRU result in performance degradation. Moreover, while some studies show success in anomaly detection, they often focus on specific applications, creating a gap in generalizable methods across diverse industries. Finally, while TinyML solutions are becoming more prevalent, their integration with advanced AI in real-time applications remains challenging due to computation, memory, and power efficiency constraints. Further research must address these issues and improve model adaptability, scalability, and performance in edge computing environments.

3. The proposed model

This work proposes a novel DLTML-RTADPM technique to secure IoT consumer devices. The major aim of the DLTML-RTADPM technique is to recognize and categorize anomalies in IoT consumer devices. To accomplish that, the DLTML-RTADPM technique has Z-score normalization, feature selection using FFA, anomaly detection using GLMS-BiLSTM, and JOA using hyperparameter tuning. Fig. 1 illustrates the entire flow of the DLTML-RTADPM technique.

3.1. TinyML deployment and development

The application development, end device sampling, and model generation are three significant steps for TinyML deployment [34]. Self-powered sensors capture the data from the IoT devices to reduce the MCU sampling rate, further reducing the power consumption. Consequently, the real-time data undergoes a few preprocessing techniques like feature selection, data normalization, and missing value imputation. The DL model is trained with the processed data and built with

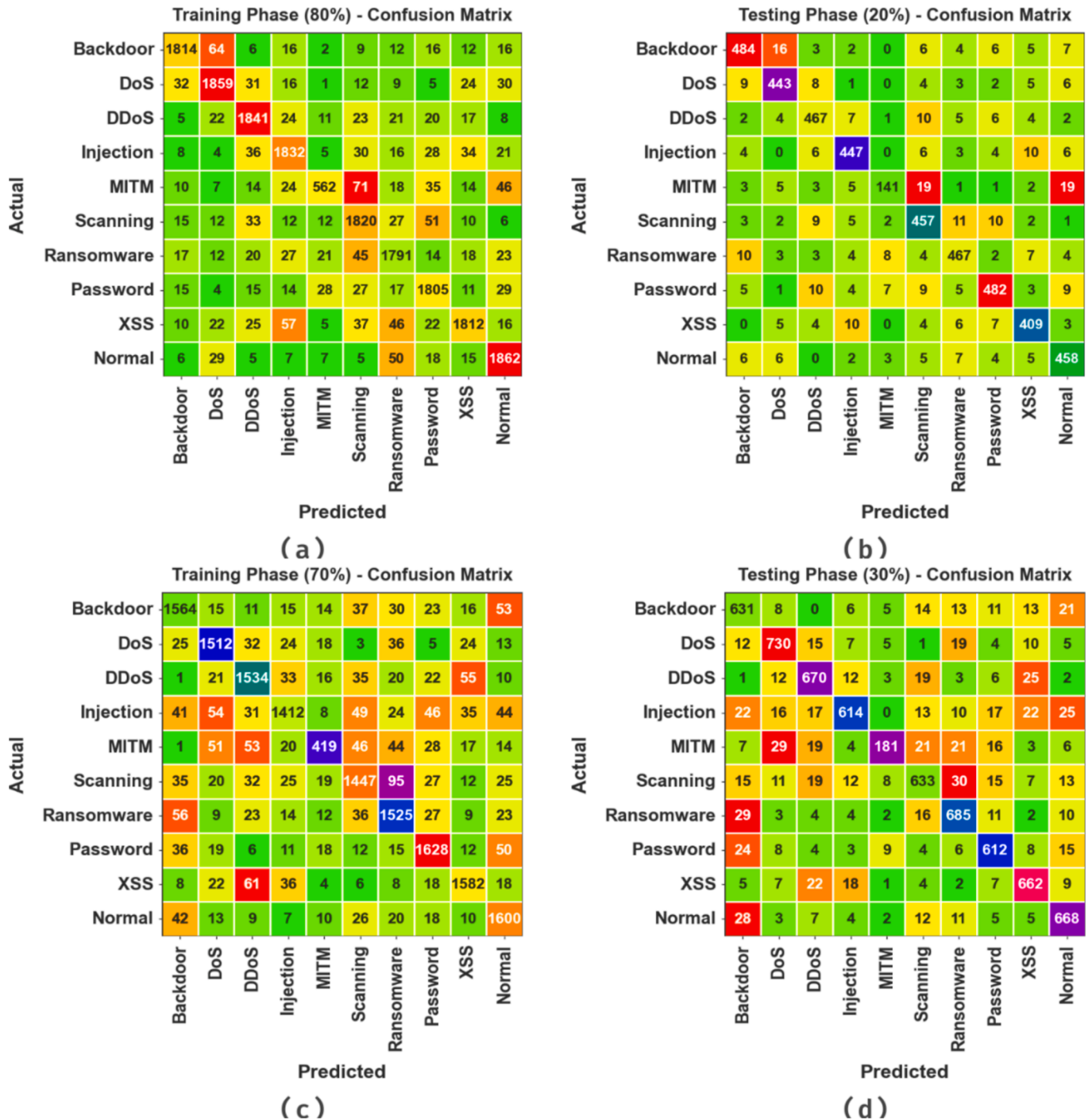


Fig. 3. Confusion matrices of (a-b) 80%TRAS/20%TESS and (c-d) 70%TRAS/ 30%TESS.

TensorFlow (Google Inc.) to obtain better accuracy in detection. Note that the smaller the model size, the lesser the inference time. TensorFlow Lite compresses this method after training; an innovative inference frame developed for running ML techniques on a resource-constraint device, with skills like pruning and parameter quantization, more minor training epochs, fewer data, and a low cost needed. Finally, the efficient models are applied to the TinyML-supported MCUs after being transformed into a suitable document for the embedded system. Various applications are customized easily due to the open and complete embedded ecosystem.

3.2. Z-score normalization

At the primary stage, the DLTML-RTADPM technique normalizes the input data using Z-score normalization [35]. This is chosen for its simplicity and efficiency in standardizing data, making it specifically appropriate for ML models that depend on the assumption of data with a normal distribution. This technique converts the data to have a mean of 0 and a standard deviation of 1, ensuring that all features contribute equally to the model's performance, regardless of their original scales. Unlike min-max scaling, Z-score normalization is less sensitive to outliers, making it robust when dealing with data containing extreme values. Additionally, it assists in accelerating the convergence of

Table 2

Attack detection outcome of DLTML-RTADPM technique with 80%TRAS and 20%TESS.

Classes	$Accu_y$	$Prec_n$	$Reca_t$	F_{Score}	$G_{Measure}$
TRAS (80 %)					
Backdoor	98.56	93.89	92.22	93.05	93.05
DoS	98.21	91.35	92.08	91.71	91.71
DDoS	98.21	90.87	92.42	91.64	91.64
Injection	97.98	90.29	90.96	90.63	90.63
MITM	98.24	85.93	70.16	77.25	77.65
Scanning	97.68	87.54	91.09	89.28	89.30
Ransomware	97.80	89.24	90.09	89.66	89.66
Password	98.04	89.62	91.86	90.73	90.73
XSS	97.90	92.12	88.30	90.17	90.19
Normal	98.21	90.52	92.91	91.70	91.71
Average	98.08	90.14	89.21	89.58	89.63
TESS (20 %)					
Backdoor	98.06	92.02	90.81	91.41	91.41
DoS	98.30	91.34	92.10	91.72	91.72
DDoS	98.15	91.03	91.93	91.48	91.48
Injection	98.32	91.79	91.98	91.88	91.88
MITM	98.32	87.04	70.85	78.12	78.53
Scanning	97.62	87.21	91.04	89.08	89.10
Ransomware	98.09	91.21	91.21	91.21	91.21
Password	97.98	91.98	90.09	91.03	91.03
XSS	98.26	90.49	91.29	90.89	90.89
Normal	97.98	88.93	92.34	90.60	90.62
Average	98.11	90.30	89.36	89.74	89.79

gradient-based optimization methods, such as in DL, and improves the accuracy of models by reducing bias caused by differing feature magnitudes.

Z-score normalization, a.k.a standardization, is a statistics model that transforms data into a uniform distribution with a mean of zero and a variance of one. This approach become more prominent in different sectors such as biology, finance, and ML to ensure that datasets with different units and scales can be effectively analyzed and compared. This method includes subtracting the mean of the dataset from the data point and then dividing by the variance. Thereby, the resultant data distribution makes it simple to analyze and interpret.

3.3. FFA-based feature selection

For the process of feature selection, the FFA is used for a high dimensionality reduction process where the optimal set of features is

chosen [36]. This approach is selected for its robust ability to handle high-dimensional data while detecting the most relevant features. Inspired by the foraging behaviour of fennec foxes, this nature-inspired method effectually navigates the feature space, balancing exploration and exploitation to avert local optima and ensure global optimization. Compared to conventional techniques such as genetic algorithms or simulated annealing, FFA improves convergence rates and robustness in selecting features, particularly in complex datasets. Its adaptive search capability enhances the quality of feature selection, resulting in enhanced model accuracy and reduced overfitting. Furthermore, the simplicity and flexibility of the FFA model make it highly appropriate for applications where computational efficiency is critical. The FFA is a new SI method stimulated by the predatory and evasion performance of the population of fennec foxes. FFA resolves optimizer problems by mimicking these two behaviours. This method primarily contains 2 phases: escaping predators and digging prey behaviours from the sand.

(1) Phase 1: Local search

During hunting, the fennec fox utilizes its sharp earshot to differentiate the prey's location in the seize, sand, and excavate the prey. Mimicking these behaviours enhances the local search capability of FFA and takes it near the optimum performance globally. N is the number of fennec foxes, i^{th} Fox's location is $X_i = (x_{i1}, x_{i2}, \dots, x_{im})$, and the fitness function under this location is F_i . A local search can be executed in the area with X_i as the centre and radius R . The location upgrade method is as:

$$x_{ij}^{p1} = x_{ij} + (2 \cdot r - 1) \cdot R_{ij} \quad (1)$$

$$R_{ij} = \alpha \cdot \left(1 - \frac{t}{T}\right) \cdot x_{ij} \quad (2)$$

$$X_i = \begin{cases} X_i^{p1}, & \text{if } F_i^{p1} < F_i(X_i) \\ \text{else} \end{cases} \quad (3)$$

whereas X_i^{p1} implies the location of i^{th} fennec fox depends on the primary stage upgrade, x_{ij}^{p1} refers to the position of its j^{th} dimensional, and F_i^{p1} denotes the FV of X_i^{p1} . R_{ij} indicates the region radius of $x_{i,j}$, t defines the iteration counter, T represents the total number of iterations, r indicates the randomly generated value between $[0,1]$, and α defines the constant fixed to 0.2.

(2) Phase 2: Global search

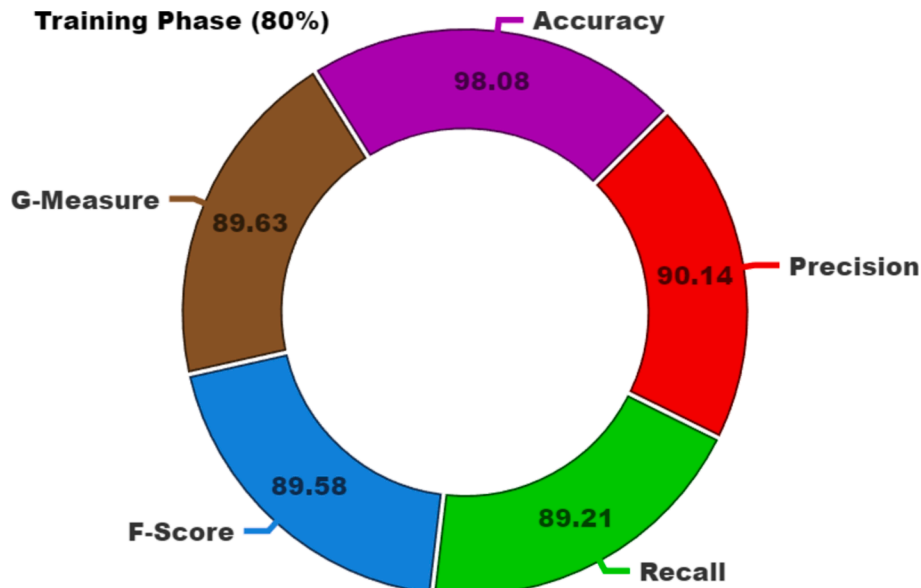


Fig. 4. Average outcome of DLTML-RTADPM technique with 80%TRAS.

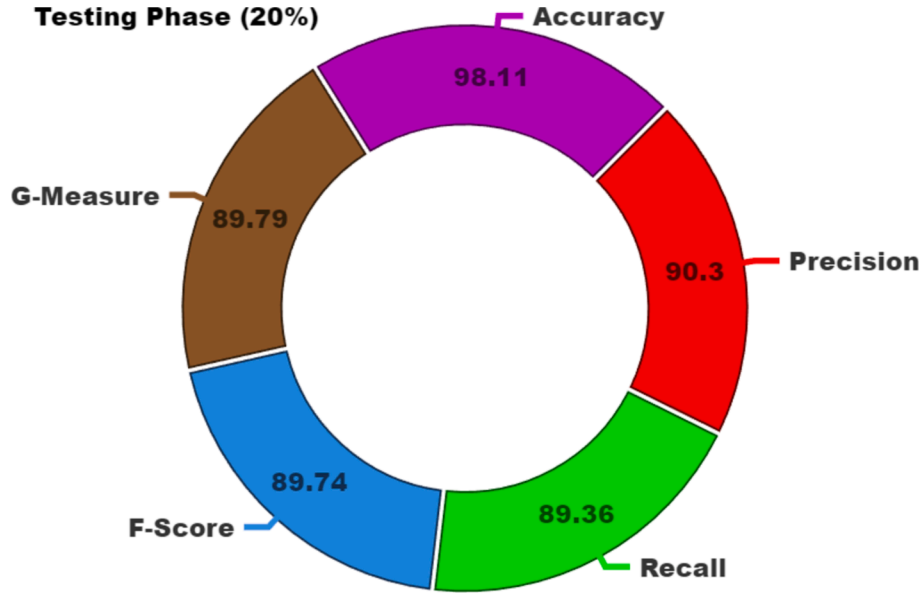


Fig. 5. Average outcome of DLTML-RTADPM technique with 20% TESS.

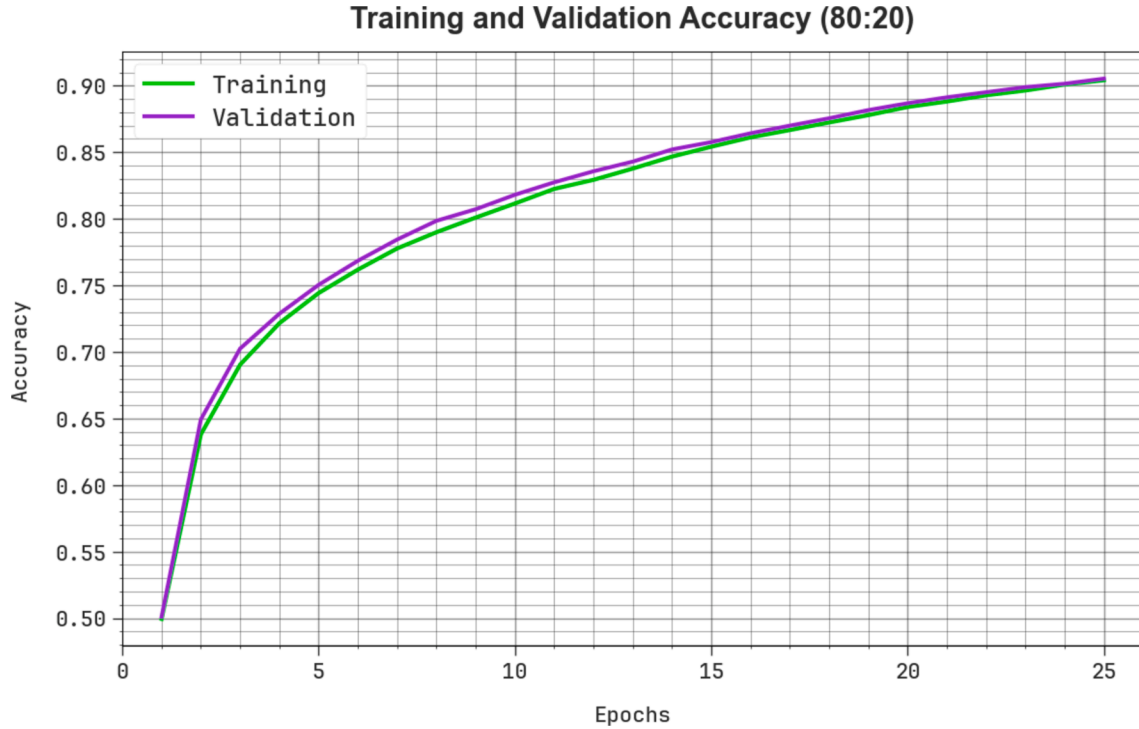


Fig. 6. Accuracy curve of dltml-rtadpm technique with 80% train and 20% test.

If encountering the predator hunt, the fennec fox escapes the predator hunts with its extraordinary speed and quick variations in path. The escaping approach of the fennec fox assists as the base for global search. These strategies improve the exploration capability, allowing it to avoid the local optimum region and recognize the global optimum region. The location upgrade method is as:

$$X_i^{rand} : x_{ij}^{rand} = x_{kj}, k \in \{1, 2, \dots, N\}, i = 1, 2, \dots, N \quad (4)$$

$$x_{ij}^{p2} = \begin{cases} x_{ij} + r \cdot (x_{ij}^{rand} - x_{ij}), & F_i^{rand} < F_i(x_{ij} + r \cdot (x_{ij}^{rand} - x_{ij})) \\ else & \end{cases} \quad (5)$$

$$X_i = \begin{cases} X_i^{p2}, & F_i^{p2} < F_j(X_i) \\ else & \end{cases} \quad (6)$$

X_i^{rand} implies the destination position of i^{th} fennec fox escaping, x_{ij}^{rand} represents the position from j^{th} dimensional, and F_i^{rand} denotes the FV at that position. X_i^{p2} stands for the position of i^{th} fennec fox depending on the secondary stage upgrade, x_{ij}^{p2} defines its position from the j^{th} dimensional, and F_i^{p2} signifies the FV at that location. r demonstrates the random integer among zero and one, and I denotes the random number chosen from the set $\{1, 2\}$.

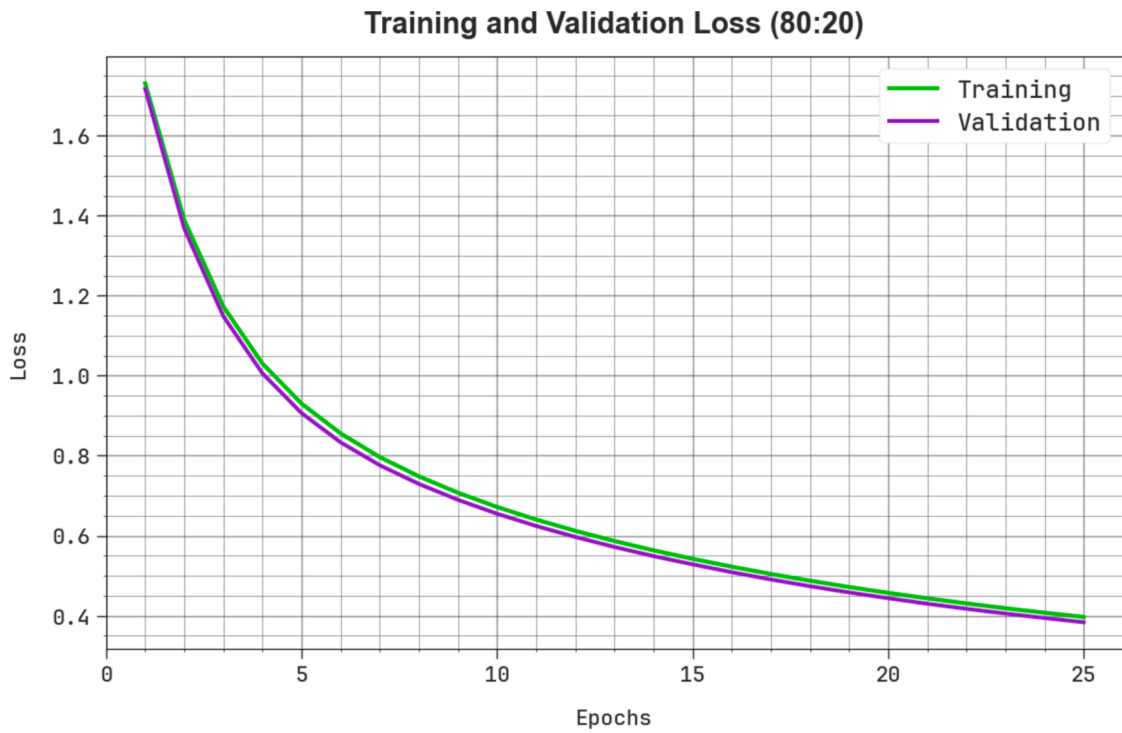


Fig. 7. Loss curve of DLTML-RTADPM method with 80%TRAS and 20%TESS.

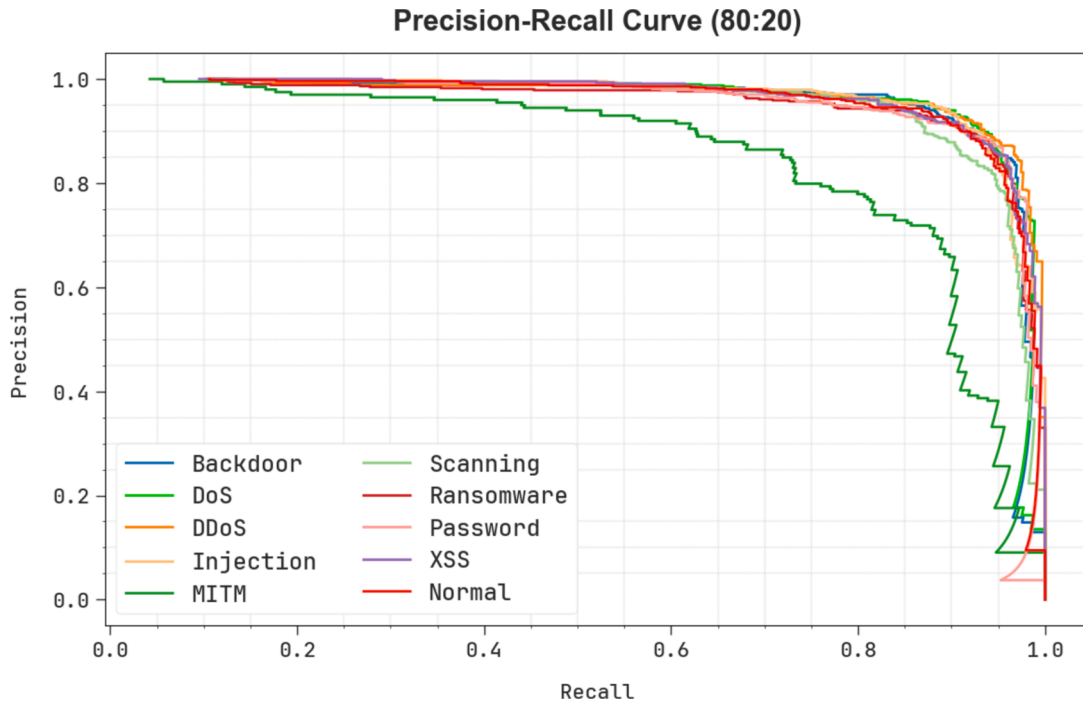


Fig. 8. PR curve of DLTML-RTADPM method with 80%TRAS and 20%TESS.

In FFA, the objectives are combined into a single formula such that the predetermined weight detects their importance [37]. Here, an FF combines both FS objectives.

$$Fitness(X) = \alpha \cdot E(X) + \beta \cdot \left(1 - \frac{|R|}{|N|}\right) \quad (7)$$

In equation (7), the fitness values of the X subset are represented as $Fitness(X)$, $E(X)$ signifies the classification rate of error over the features

selected in the subset X , $|R|$ and $|N|$ indicate the number of features chosen and the number of real features in the dataset correspondingly, and the weights of the reduction ratio and the classifier error are α and β , $\alpha \in [0, 1]$ and $\beta = (1 - \alpha)$.

3.4. Anomaly detection using GLMS-BiLSTM

The DLTML-RTADPM technique utilizes the GLMS-BiLSTM model for

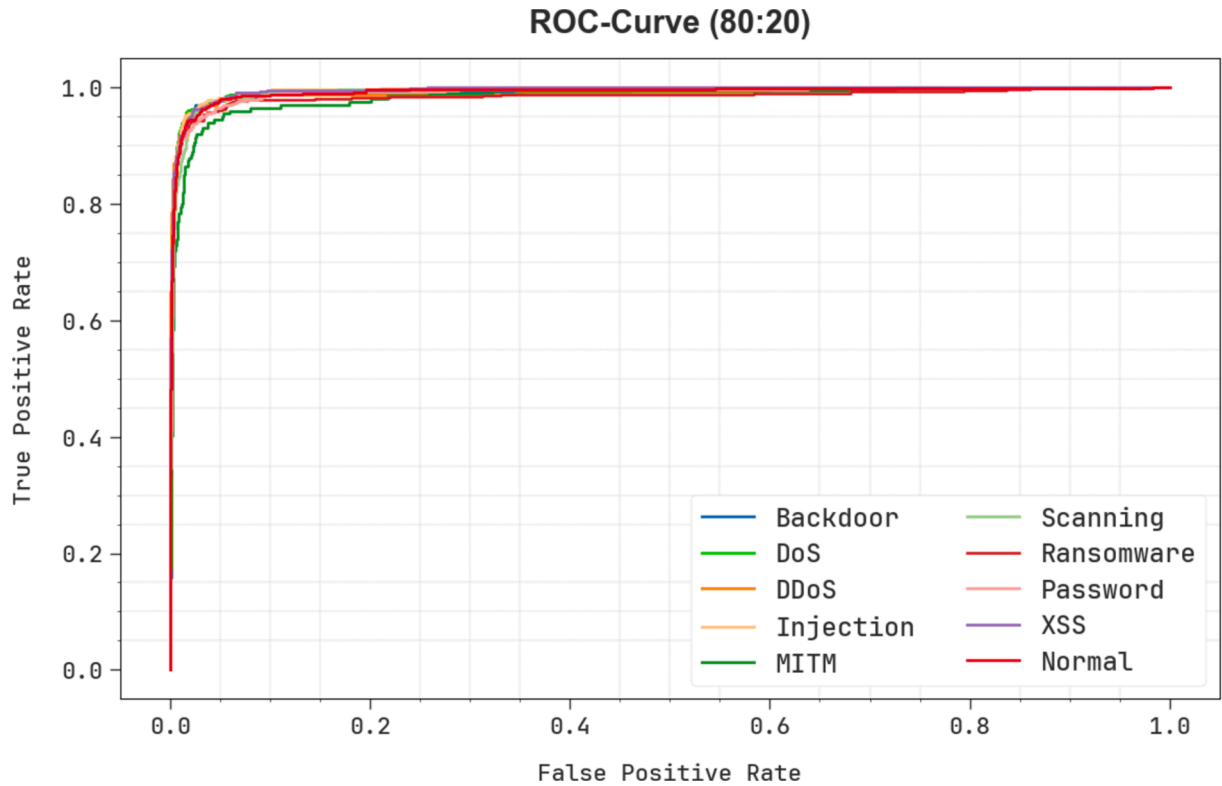


Fig. 9. ROC curve of DLTML-RTADPM technique with 80%TRAS and 20%TESS.

Table 3
Attack detection outcome of DLTML-RTADPM technique with 70%TRAS and 30%TESS.

Classes	Accu _y	Prec _n	Rec _{a_i}	F _{Score}	G _{Measure}
TRAS (70 %)					
Backdoor	97.21	86.46	87.96	87.20	87.21
DoS	97.54	87.10	89.36	88.21	88.22
DDoS	97.14	85.60	87.81	86.69	86.70
Injection	96.86	88.42	80.96	84.53	84.61
MITM	97.61	77.88	60.46	68.07	68.62
Scanning	96.72	85.27	83.30	84.27	84.28
Ransomware	96.95	83.93	87.95	85.89	85.91
Password	97.61	88.38	90.09	89.23	89.23
XSS	97.74	89.28	89.73	89.50	89.51
Normal	97.54	86.49	91.17	88.77	88.80
Average	97.29	85.88	84.88	85.24	85.31
TESS (30 %)					
Backdoor	96.68	81.52	87.40	84.36	84.41
DoS	97.52	88.27	90.35	89.30	89.30
DDoS	97.30	86.23	88.98	87.58	87.59
Injection	96.99	89.77	81.22	85.28	85.38
MITM	97.72	83.80	58.96	69.22	70.29
Scanning	96.68	85.89	82.96	84.40	84.41
Ransomware	97.22	85.62	89.43	87.48	87.50
Password	97.55	86.93	88.31	87.62	87.62
XSS	97.59	87.45	89.82	88.62	88.63
Normal	97.40	86.30	89.66	87.95	87.97
Average	97.27	86.18	84.71	85.18	85.31

anomaly detection [38]. This methodology is chosen for its ability to capture both past and future dependencies in time-series data, which is significant for detecting anomalies in dynamic systems. The BiLSTM model improves the method's performance by processing the data in both forward and backward directions, allowing it to understand the temporal context better. Combined with GLMS, this model presents efficient and adaptive learning, making it appropriate for scenarios where real-time anomaly detection is required. Compared to

conventional methods, GLMS-BiLSTM can efficiently handle non-linear relationships and complex data patterns while providing faster convergence and lower computational costs. Its flexibility in adjusting to evolving data distributions makes it a robust choice for diverse anomaly detection tasks. Fig. 2 demonstrates the structure of BiLSTM.

In this work, *i* introduce an evolutionary solution for anomaly detection in metallurgical facilities embodied in the Bi-LSTM that integrates GLMS. This innovative methodology is adept at conducting a comprehensive review of vibrational data, enabling specific anomaly detection. Incorporating the GLMS technique considerably refines the learning process, thereby improving the accuracy and efficiency of anomaly prediction. This novel approach is a powerful tool to ensure metallurgical industry security and enhance anomaly detection performance. It had been a milestone in improving the reliability of equipment and the efficiency of the production process. This represents the effective incorporation of cutting-edge ML techniques with real-time engineering applications, offering an effective tool for operational optimization and maintenance in heavy industries and setting new standards in the mechanical anomaly diagnoses field. The GLMS-BiLSTM model applies a Bi-LSTM framework in managing vibration data, which is adept at encapsulating the nuances of time series data. This architecture includes cell state, input, forget, and output gates and plays a major role in the model functionality.

Initially, the input gate defines which anomaly information should be updated. This gate includes a tanh layer that makes original candidate values, which are added to the cell state, and a sigmoid layer that defines what value requires updating.

$$i_t = \sigma(w_{xi}x_t + w_{hi}h_{t-1} + b_i) \quad (8)$$

$$\tilde{C}_t = \tanh(w_{xc}x_t + w_{hc}h_{t-1} + b_c) \quad (9)$$

Next, the gate of forgetting defines what data is forgotten from the cell state. This scrutinizes h_{t-1} and x_t via a sigmoid layer and outputs a number within [0, 1] for the cell state C_{t-1} . The value of 0 specifies complete discard, whereas 1 indicates complete retention.

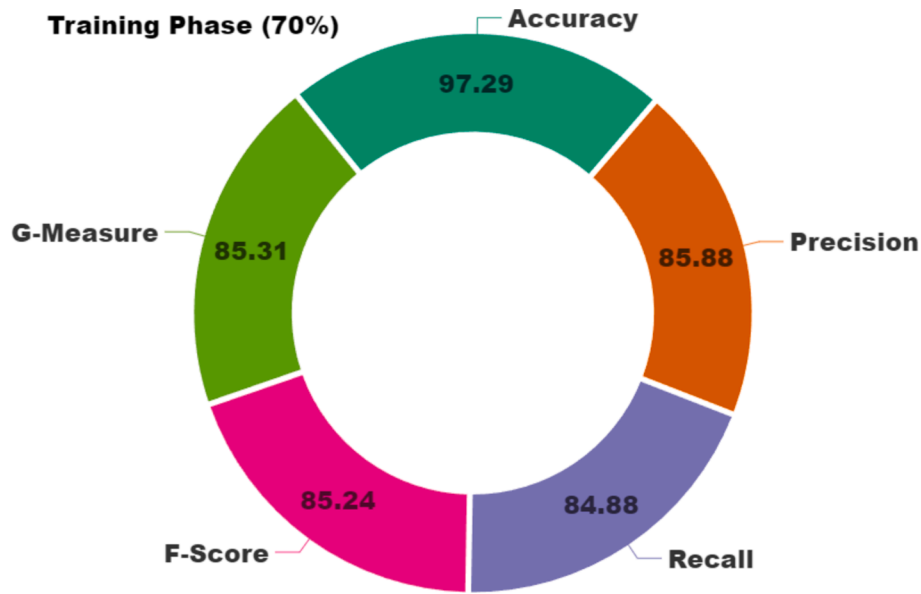


Fig. 10. Average outcome of DLTML-RTADPM technique with 70%TRAS.

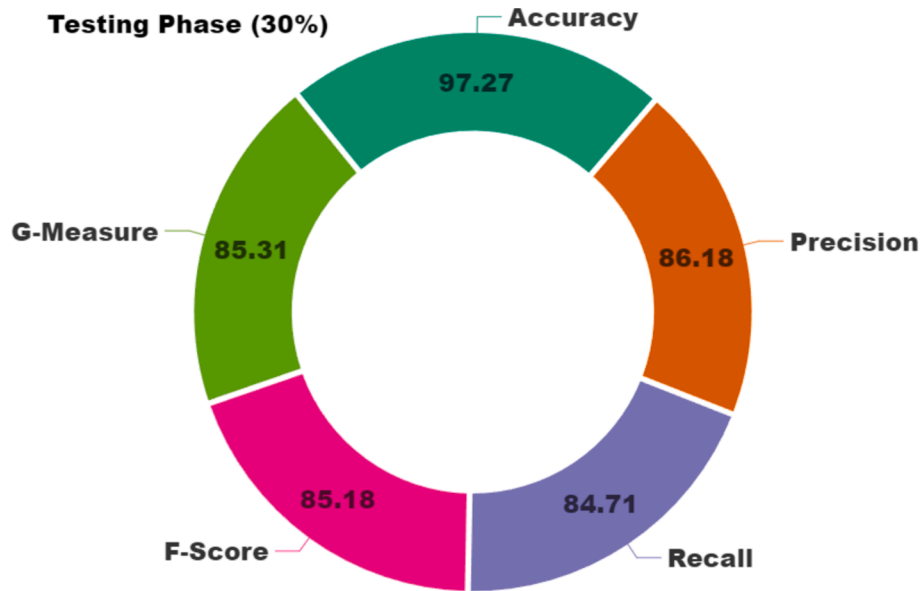


Fig. 11. Average outcome of DLTML-RTADPM technique with 30%TESS.

$$f_t = \sigma(w_{xf}x_t + w_{hf}h_{t-1} + b_f) \quad (10)$$

This method updates the cell state by integrating the data from the forget and input gates, storing and filtering relevant data.

$$C_t = f_t \bullet C_{t-1} + i_t \bullet \tilde{C}_t \quad (11)$$

Lastly, the output gate defines the value of the next hidden state. This comprises data from the preceding time step and is used for the prediction.

$$o_t = \sigma(w_{xo}x_t + w_{ho}h_{t-1} + b_o) \quad (12)$$

$$h_t = o_t \bullet \tanh(C_t) \quad (13)$$

Here, $\tanh$ and σ are the hyperbolic tangent functions and sigmoid, $\bullet$, entry-wise multiplication. W and b represent the weight matrix and biased vector correspondingly. C_t denotes the cell state. The input x_t

represents the existing time step, and h_t shows the hidden state. The model efficiently captures the short—and long-range dependencies in time sequence data, which makes it suitable for data processing tasks.

The GLMS technique introduces a new solution for time sequence data processing with noise and its high robustness to noisy data, integrated with the unique capabilities of BiLSTM to capture long-range dependency. This method improves the prediction method's stability and precision and improves the network's adaptive ability in dynamic environments. The GLMS technique enhances the weight and bias of the BiLSTM technique by minimizing the prediction of mean square error. It exploits the gradient descent optimization to calculate the gradient of prediction error, which consequently adjusts the network parameter. This effectively increases the overall capacity of the predictive model and decreases the error margins.

In the GLMS-BiLSTM network, the optimization of Bi-LSTM using GLMS often entails two significant steps:

First, the error between the actual value and the current prediction is

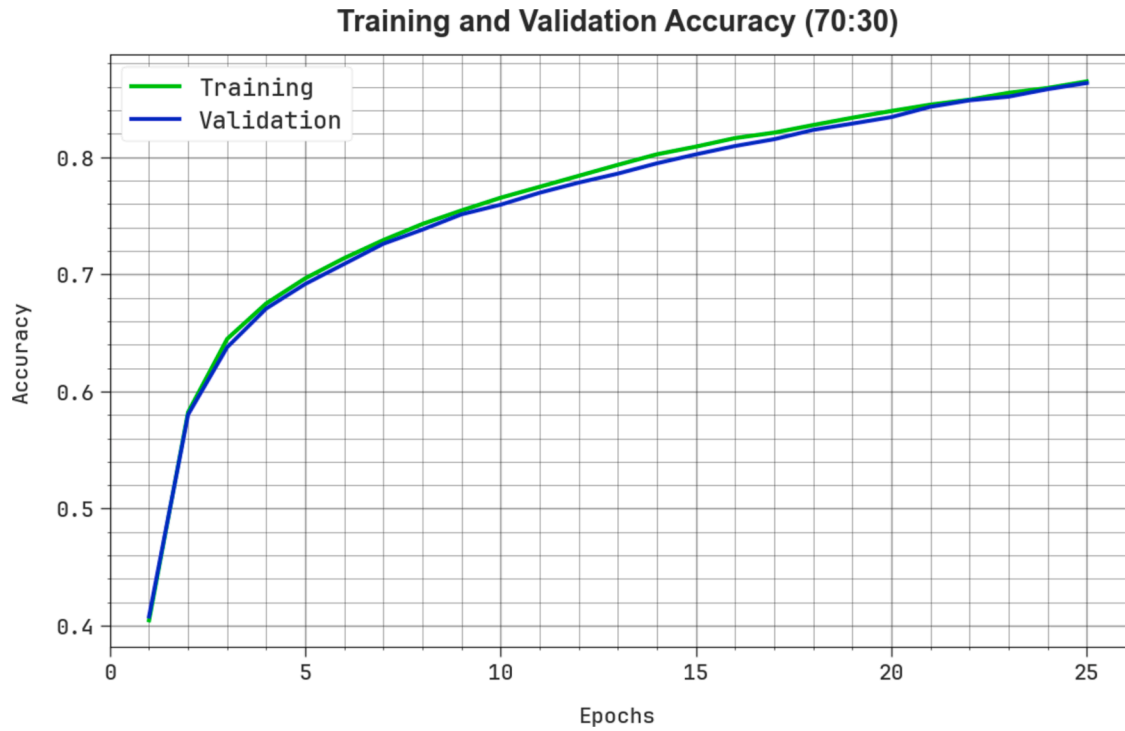


Fig. 12. Accu curve of dltml-rtadpm technique with 70%tras and 30%tess.

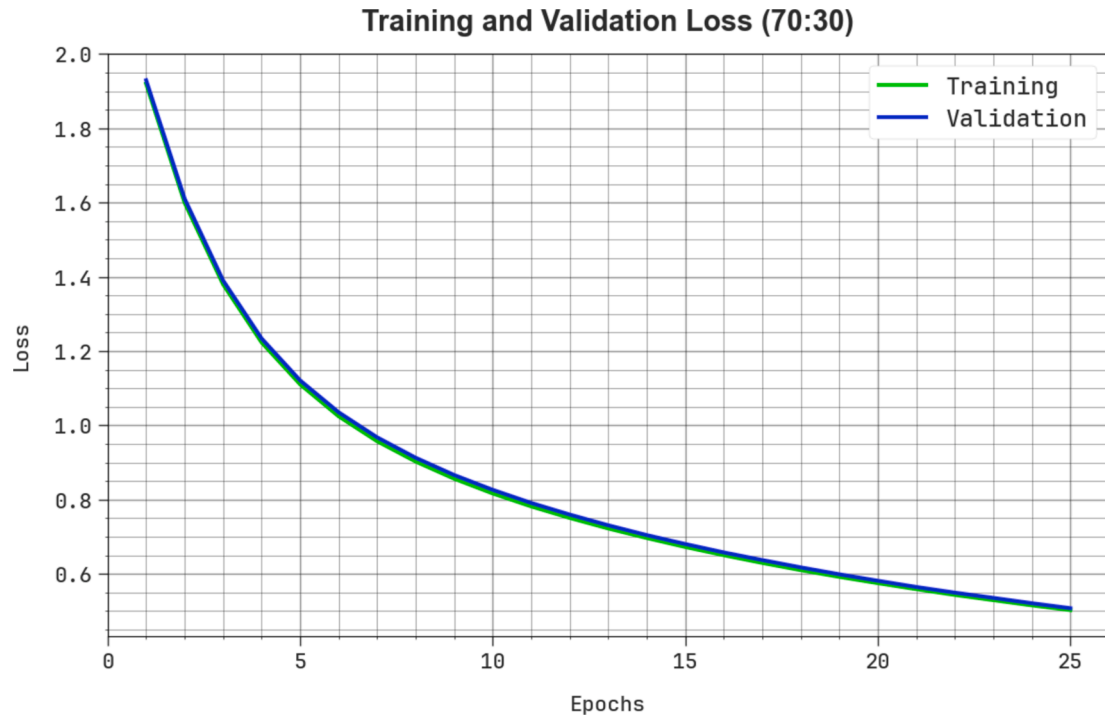


Fig. 13. Loss curve of DLTML-RTADPM method with 70%TRAS and 30%TESS.

computed using the following equation.

$$e(t) = d(t) - w^T(t) \cdot x(t) \quad (14)$$

where $e(t)$ refers to the error at time t , $d(t)$ denotes the actual value, $w(t)$ indicates the weight vector, and $x(t)$ shows the input feature vector.

Next, the weight is updated based on the error gradient as follows:

$$w(t+1) = w(t) + \mu \cdot x(t) \cdot e(t) \quad (15)$$

In Eq. (15), μ denotes the learning rate.

The above steps strengthen the GLMS-BiLSTM network's optimization strategy, which accurately calibrates the Bi-LSTM model for precise and efficient anomaly detection in mechanical systems. By constantly changing the weight regarding the computed error gradient, the model efficiently improves its accuracy and reliability in anomaly diagnosis and minimizes prediction errors. This optimization technique underscores its fitness for real-time, complex applications in mechanical

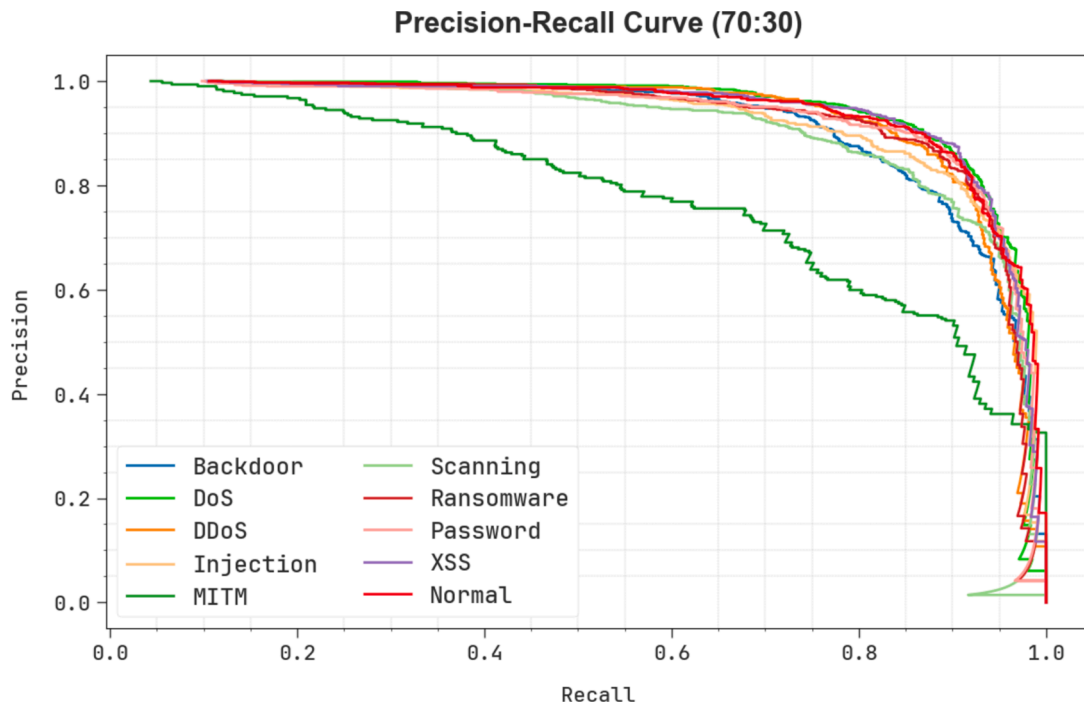


Fig. 14. PR curve of DLTML-RTADPM method with 70%TRAS and 30%TESS.

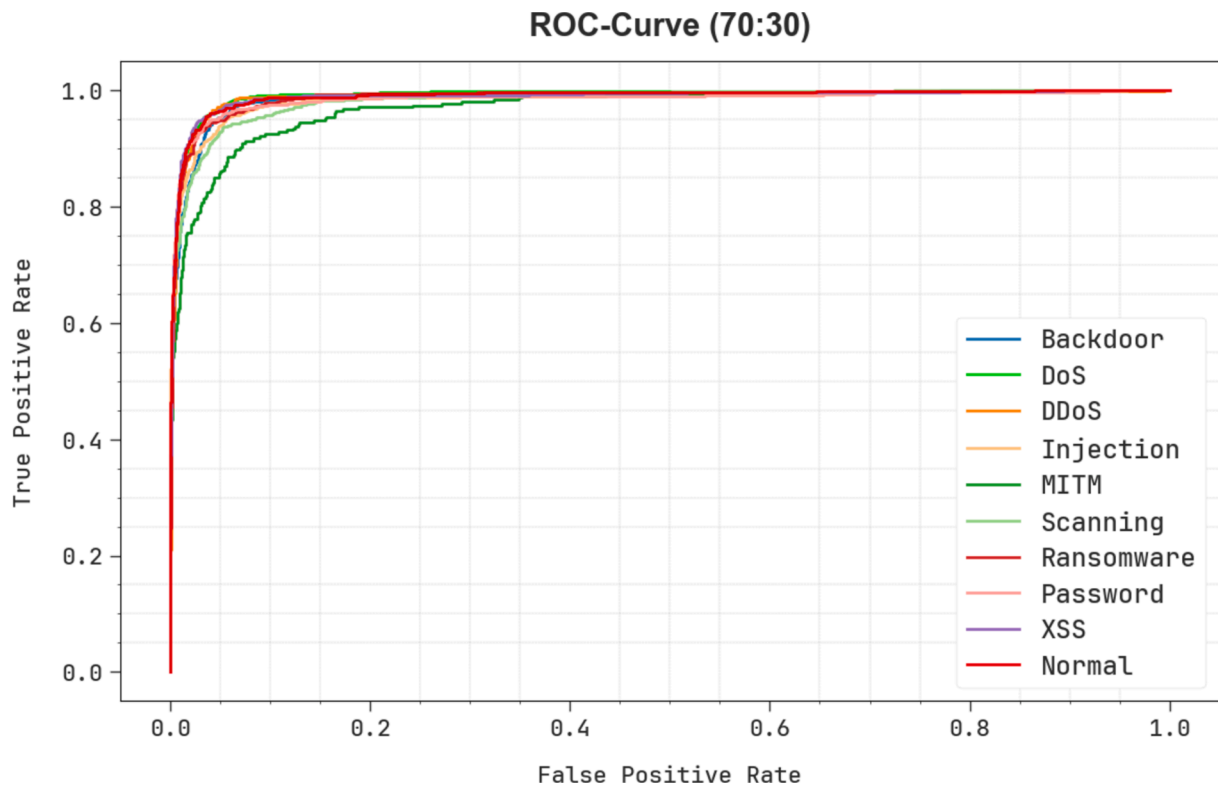


Fig. 15. ROC curve of DLTML-RTADPM technique with 70%TRAS and 30%TESS.

anomaly diagnosis and reflects the complexity of the GLMS-BiLSTM network.

3.5. Hyperparameter tuning using JOA

The JOA-based hyperparameter tuning process is involved in

improving further the detection results of the DLTML-RTADPM technique [39]. This model is chosen due to its simplicity, ease of implementation, and efficiency in optimizing complex models without needing gradient information. Unlike conventional optimization techniques, JOA does not depend on prior knowledge of the problem's landscape, making it highly adaptable to various ML tasks. Its

Table 4

Comparative analysis of DLTML-RTADPM technique with other models [41–44].

Techniques	Accu _y	Prec _n	Reca _i	F1 _{Score}
LSTM Model	90.94	82.21	82.15	85.39
GADAD-RF	94.00	86.36	85.60	88.14
GADAD-ET	95.00	83.53	87.02	82.39
GA-RF Model	94.00	83.34	88.10	82.60
DT	97.67	83.30	82.01	89.28
LR	96.69	89.21	83.04	82.55
GB	97.50	84.26	88.12	82.92
CNN-LSTM	91.74	82.80	82.95	85.47
GRU-DAE	94.62	86.96	86.27	88.21
SAAE-DNN	95.80	84.32	87.60	82.46
GoogleLeNet	94.71	83.86	88.70	82.66
SDAE-ELML	97.37	83.90	82.61	88.33
DLTML-RTADPM	98.11	90.30	89.36	89.74

population-based search ensures diversity, preventing local minima and improving the likelihood of finding the global optimum. Additionally, the capability of the JOA model to dynamically adjust search parameters improves convergence speed and accuracy, making it more effectual than other methods, such as grid or random search. Its robustness in handling large search spaces and averting overfitting enhances model performance and generalization.

It is a development in the domain of population-based *meta*-heuristic optimizer method by R. Venkata Rao at the year of 2015. The word JAYA comes from the Sanskrit language, which indicates victory. This can be stimulated through the social activities of the population. Similarly, the population size denotes the count of functions, and every unit in the population signifies the tasks and final condition included. At all rounds, all particles retain on altering their position in the difficult space and attempt to better optimum solution. The principle of such a method needs considerably fewer control parameters such as population size, count of design variables, and maximum number of generations. This describes other present evolutionary methods or some nature-based techniques because of their computational efficiency. The effectiveness of this method has been computed by a context-specific fitness function (FF). Particles' fitness values are measured by applying FF, and the worst and best results have been accordingly recognized. Therefore, the fitness of all particles in the population than prior fitness values in every production to find the best place and renovated through Eq. (16). In the

formula, two random factors, r_1 and r_2 , have been applied as control variables for generating a balance among global and local searches. The solution could be changed depending on the formula's best and worst fitness values.

$$X_i^{k+1} = X_i^k + r_1(p_{best_i} - |X_i^k|) - r_2(p_{worst_i} - |X_i^k|) \quad (16)$$

Here, the specified equation can define the particle's initial location.

$$X_0^k = X_{min} + (X_{max} - X_{min}) \times r \quad (17)$$

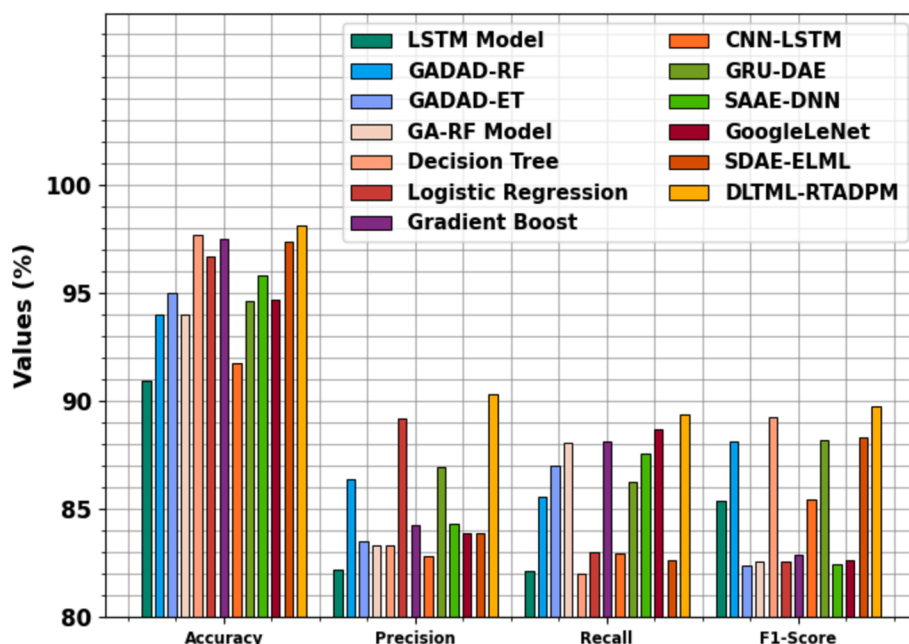
X_i^k is the existing location of i^{th} particles at iteration k , X_i^{k+1} refers to the updated values of i at iteration $k+1$, and r_1 and r_2 have two random numbers ranging among (0, 1). This form $r_1(p_{best_i} - |X_i^k|)$ specifies the trend of the solution for moving nearby towards the optimum solution, and $-r_2(p_{worst_i} - |X_i^k|)$ shows the trend of the decision to leak the poor solution. This solution has been known when the value of X_i^{k+1} can have improved fitness values than X_i^k , $X_{min} = -0.4$, $X_{max} = 4.0$, and $r = 0.1$. The X_{max} value has been set to 4.0 since it could be limited particles that prevent the particles from flying out unpredictably and converging rapidly without exploring.

In all repetition, the population acquires a novel position, and such a

Table 5

CT analysis of DLTML-RTADPM technique with other models.

Techniques	CT (sec)
LSTM Model	18.80
GADAD-RF	19.11
GADAD-ET	20.01
GA-RF Model	17.29
DT	20.46
LR	15.63
GB	18.40
CNN-LSTM	20.18
GRU-DAE	18.23
SAAE-DNN	18.30
GoogleLeNet	20.78
SDAE-ELML	16.96
DLTML-RTADPM	10.60

**Fig. 16.** Comparative analysis of DLTML-RTADPM technique with other approaches.

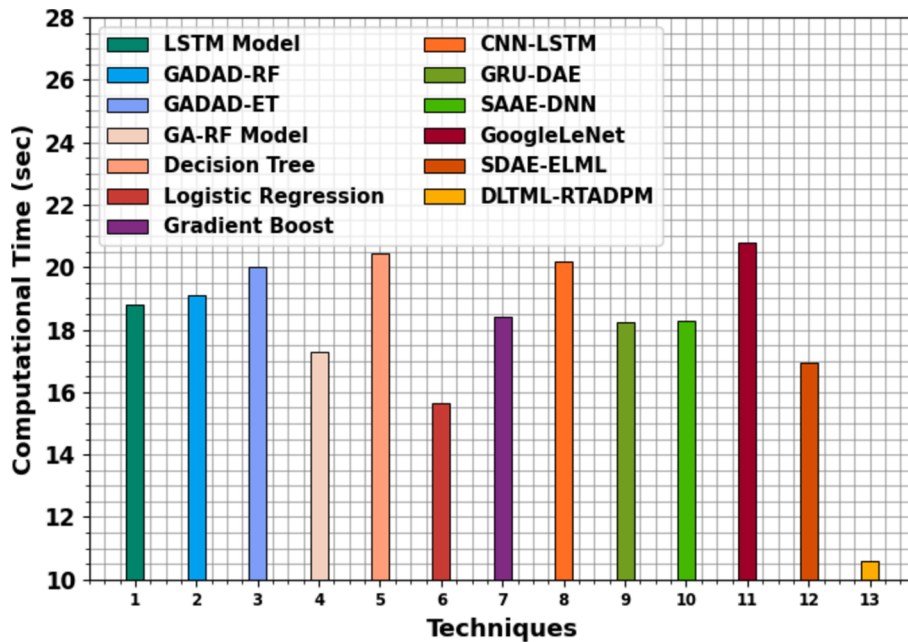


Fig. 17. CT analysis of DLTML-RTADPM technique with other approaches.

method must be repeated till the best and worst places cannot be modified. Due to the ease of performance, this method provides lower time and computation complexities as the outcomes have been improved in a single stage with the help of a single formula. This method exits the local optimum successfully with a higher convergence rate as these two best and worst solutions must be deliberated in a single calculation. The user has not recognized the difficulties in tuning the particular method factors for the tasks as this technique could not be needed for a few of these parameters except random numbers.

The JOA derives the FF to gain superior classification outcomes. It defines the positive integer to describe the high efficiency of the solution candidate. This study observes a decrease in the classifier error rate for the FF.

$$fitness(x_i) = ClassifierErrorRate(x_i) = \frac{No.ofmisclassifiedsamples}{TotalNo.ofsamples} \times 100 \quad (18)$$

4. Result analysis and discussion

The DLTML-RTADPM technique's experimental analysis is studied on the benchmark ToN-IoT dataset [40]. It includes 23,500 instances with 10 class labels, as illustrated in Table 1.

Fig. 3 demonstrates the confusion matrices produced by the DLTML-RTADPM approach in terms of 70 %TRAS/ 30 %TESS and 80 %TRAS/ 20 %TESS. The results indicate that the DLTML-RTADPM model effectively classified and recognized each 10-number class.

The attack recognition values of the DLTML-RTADPM approach with 80 %TRAS and 20 %TESS are described in Table 2. Fig. 4 demonstrates the average results offered by the DLTML-RTADPM approach on 80 % TRAS. These values exhibit that the DLTML-RTADPM approach accomplished efficient identification of class labels. It is also noticed that the DLTML-RTADPM model gains an average $accu_y$ of 98.08 %, $prec_n$ of 90.14 %, $reca_l$ of 89.21 %, F_{score} of 89.58 %, and $G_{measure}$ of 89.63 %.

Fig. 5 demonstrates the average results offered by the DLTML-RTADPM technique on 20 %TESS. These values exhibit that the DLTML-RTADPM technique accomplished effective identification of class labels. The DLTML-RTADPM methodology also gains an average $accu_y$ of 98.11 %, $prec_n$ of 90.30 %, $reca_l$ of 89.36 %, F_{score} of 89.74 %, and $G_{measure}$ of 89.79 %.

The classifier results of the DLTML-RTADPM model are established in Fig. 6 in the VALA and TRAA investigation method with 20 %TESS and 80 %TRAS. This figure shows an essential understanding of the performance of the DLTML-RTADPM approach across several epochs, demonstrating its process of learning and generalization abilities. Notably, the figure designates a steady growth in the VALA and TRAA with improving epoch counts. It guarantees the adaptive nature of the DLTML-RTADPM method in the pattern recognition technique on both databases. This growing trend in VALA defines the proficiency of the DLTML-RTADPM approach to familiarize the TRA data and outshine in producing a precise classifier of unseen data, indicating strong generalizability.

Fig. 7 depicts a wide-ranging analysis of the VALL and TRLA consequences of the DLTML-RTADPM technique across various numbers of epochs with 20 %TESS and 80 %TRAS. The gradual decrease in TRLA underlines the DLTML-RTADPM methodology, which improves the weightings and decreases the classification error on both databases. This figure accurately explains the DLTML-RTADPM model's relationship with the TRA datasets, underscoring the capacity to capture patterns inside both databases. Notably, the DLTML-RTADPM technique consistently improves its parameters in minimizing the differences between the prediction and actual TRA class labels.

Reviewing the PR investigation, as exhibited in Fig. 8, the values guarantee that the DLTML-RTADPM model increasingly gains improved PR values under every class with 80 %TRAS and 20 %TESS. This confirms the improved proficiencies of the DLTML-RTADPM methodology in the detection of different numbers of classes, presenting ability in the detection of class labels.

Moreover, in Fig. 9, the examination of ROC produced in the DLTML-RTADPM model outperformed the classifier of distinct labels with 80 % TRAS and 20 %TESS. It thoroughly explains the tradeoff between FRP and TPR above different numbers of epochs and recognition threshold values. This figure emphasizes the better classifier consequences of the DLTML-RTADPM approach in each number of classes and outlines the proficiency in addressing several classifier difficulties.

The attack detection outcomes of the DLTML-RTADPM method with 30 %TESS and 70 %TRAS are demonstrated in Table 3. Fig. 10 illustrates the average values provided by the DLTML-RTADPM model on 70 % TRAS. These performances display that the DLTML-RTADPM approach obtained efficient detection of classes. The DLTML-RTADPM method

obtains an average acc_{ty} of 97.29 %, $prec_n$ of 85.88, $reca_l$ of 84.88 %, F_{score} of 85.24 %, and $G_{measure}$ of 85.31 %.

Fig. 11 illustrates the average outcomes provided by the DLTML-RTADPM technique on 30 %TESS. These results show that the DLTML-RTADPM methodology obtained efficient detection of classes. Note that the DLTML-RTADPM technique obtains an average acc_{ty} of 97.27 %, $prec_n$ of 86.18, $reca_l$ of 84.71 %, F_{score} of 85.18 %, and $G_{measure}$ of 85.31 %.

The classifier results of the DLTML-RTADPM approach are depicted in Fig. 12 in the VALA and TRAA investigation method with 30 %TESS and 70 %TRAS. This figure shows an essential understanding of the performance of the DLTML-RTADPM approach across different epochs, demonstrating its learning and generalization abilities. Notably, the figure designates a steady growth in the VALA and TRAA with improving epoch counts. It guarantees the adaptive nature of the DLTML-RTADPM method in the pattern recognition technique on TRA and TES databases. This growing trend in VALA illustrates the proficiency of the DLTML-RTADPM methodology to adjust to the TRA data and outshine in giving precise classifiers on unseen datasets, demonstrating strong generalizability.

Fig. 13 represents a detailed examination of the VALL and TRLA values of the DLTML-RTADPM technique across various numbers of epochs with 30 %TESS and 70 %TRAS. The gradual decline in TRLA underscores the DLTML-RTADPM approach, increasing weightings and decreasing classifier errors on both databases. This figure establishes a strong understanding of the DLTML-RTADPM approach's overtone through the TRA datasets, which underlines the capacity to take patterns inside both databases. Notably, the DLTML-RTADPM approach continuously improves its parameters to minimize the changes between the prediction and real TRA classes.

Examining the PR examination, as exhibited in Fig. 14, the consequences guaranteed that the DLTML-RTADPM technique increasingly gains improved PR values under every class with 70 %TRAS and 30 % TESS. It proves the DLTML-RTADPM approach has better abilities in detecting different numbers of classes, exhibiting efficiency in detecting class labels.

Moreover, in Fig. 15, the ROC investigation produced by the DLTML-RTADPM model outshined the classifier of different labels with 70 % TRAS and 30 %TESS. It comprehensively explains the tradeoff between FRP and TPR across various numbers of epochs and recognition threshold values. This figure emphasizes the enhanced classifier performances of the DLTML-RTADPM approach in each class and outlines its effectiveness in addressing several classifier difficulties.

The DLTML-RTADPM technique's experimental outcomes are compared with other methods in Table 4 and Fig. 16 [41–44]. The result is simply that the LSTM model exhibits the least results. Next to that, the GADAD-RF, GADAD-RT, and GA-RF models have shown slightly boosted performance. Meanwhile, the Decision Tree (DT), Logistic Regression (LR), and Gradient Boosting (GB) models have managed to display considerable outcomes. Furthermore, the CNN-LSTM, Gated Recurrent Unit with Denoising Autoencoder (GRU-DAE), Stacked Autoencoder and Deep Neural Network (SAAE-DNN), GoogleLeNet, and Stacked Denoising Autoencoder with Extreme Learning Machine (SDAE-ELML) models attained slightly moderate values. But the DLTML-RTADPM technique shows its supremacy with maximum acc_{ty} , $prec_n$, $reca_l$, and $F1_{score}$ of 98.11 %, 90.30 %, 89.36 %, and 89.74 %, correspondingly. Thus, the DLTML-RTADPM method is subjugated for improved detection performance.

Table 5 and Fig. 17 illustrate the computational time (CT) analysis of the DLTML-RTADPM technique for the various methods presented in the table, reflecting the efficiency of each model in terms of processing speed. Among the methods, the DLTML-RTADPM model illustrates the fastest performance with a CT of 10.60 s, significantly outperforming other approaches. For instance, the LSTM model, GADAD-RF, and CNN-LSTM exhibit times around 18–20 s, while techniques such as DT and LR are also relatively fast, with times around 15–20 s. On the contrary,

more complex models like GoogleLeNet and GADAD-ET take longer to process, with times surpassing 20 s. The outcomes highlight that while models like DLTML-RTADPM are computationally effectual, a balance between model complexity and processing time must be considered based on the specific application needs.

5. Conclusion

This work introduced a new DLTML-RTADPM method to secure IoT consumer devices. The DLTML-RTADPM method aimed to recognize and categorize the anomalies in IoT consumer devices. To accomplish that, the DLTML-RTADPM technique has Z-score normalization, feature selection using FFA, anomaly detection using GLMS-BiLSTM, and JOA using hyperparameter tuning. At the preliminary stage, the DLTML-RTADPM method normalized the input dataset using Z-score normalization. In the DLTML-RTADPM method, the FFA was used for a high dimensionality reduction process where the optimal feature set is chosen. For anomaly detection, the DLTML-RTADPM technique utilized the GLMS-BiLSTM model. The JOA-based hyperparameter tuning process was used further to increase the detection results of the DLTML-RTADPM technique. A series of simulations were performed on the benchmark dataset to ensure better detection outcomes of the DLTML-RTADPM model. The investigational validation of the DLTML-RTADPM method portrayed a superior accuracy value of 98.11 % over other techniques. The limitations of the DLTML-RTADPM method include reliance on a specific dataset, which may limit the model's generalizability to different types of data or domains. Additionally, the performance of the model could degrade with noisy or unbalanced data, as it may not be robust to such discrepancies. The computational complexity of the proposed approach may also pose challenges when scaling to massive datasets or real-time applications. Moreover, the model's sensitivity to hyperparameters and tuning may need substantial computational resources and time. Future work may focus on improving the robustness of the model to noisy data, improving scalability, and exploring its applicability across diverse domains. Additionally, investigating more effectual optimization methods could assist in mitigating computational costs.

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CRediT authorship contribution statement

Iyad Katib: Writing – original draft, Investigation, Conceptualization, Data curation, Supervision. **Emad Albassam:** Investigation, Resources, Visualization, Writing – review & editing. **Sanaa A. Sharaf:** Software, Resources, Investigation, Validation, Writing – review & editing. **Mahmoud Ragab:** Project administration, Conceptualization, Formal analysis, Methodology, Writing – original draft.

Ethics approval

This article does not contain any studies with human participants performed by any of the authors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data supporting this study's findings are openly available at <https://research.unsw.edu.au/projects/toniot-datasets>, [40].

References

- [1] Siddique T, Mahmud MS. Physics-enhanced TinyML for real-time detection of ground magnetic anomalies. *IEEE Access* 2024.
- [2] Chaudhary G, Srivastava S, Khari M. Generative edge intelligence for securing IoT-assisted smart grid against cyber-threats. *Int J Wireless & Ad Hoc Commun* 2023;6 (1).
- [3] Khan MW, Obaidat MS, Mahmood K, Batool D, Badar HMS, Aamir M, et al. Real-time road damage detection and infrastructure evaluation leveraging unmanned aerial vehicles and tiny machine learning. *IEEE Internet Things J* 2024.
- [4] Bagheri M, Farshforoush N, Bagheri K, Shemirani AI. Applications of Artificial Intelligence Technologies in Water Environments: From Basic Techniques to Novel Tiny Machine Learning Systems. *Process Safety and Environmental Protection*. 2023.
- [5] Antonini M, Pincheira M, Vecchio M, Antonelli F. Tiny-MLOps: A framework for orchestrating ML applications at the far edge of IoT systems. In 2022 IEEE International Conference on evolving and adaptive intelligent systems (EAIS) (pp. 1-8). IEEE; 2022, May.
- [6] Kharche A, Badholia S, Upadhyay RK. Implementation of blockchain technology in integrated IoT networks for constructing scalable ITS systems in India. *Blockchain: Research and Applications*; 2024, p.100188.
- [7] Kontogiannis S, Gkamas T, Pikridas C. Deep learning stranded neural network model for the detection of sensory triggered events. *Algorithms* 2023;16(4):202.
- [8] Zhang Y, Suleiman B, Alibasa MJ, Farid F. Privacy-aware anomaly detection in IoT environments using fedgroup: a group-based federated learning approach. *J Netw Syst Manag* 2024;32(1):20.
- [9] Hammad SS, Iskandaryan D, Trilles S. An unsupervised TinyML approach is applied to the detection of urban noise anomalies under the smart cities environment. *Internet Things* 2023;23:100848.
- [10] tu Zahra F, Bostanci YS, Soyuturk M. LSTM-based jamming detection and forecasting model using transport and application layer parameters in Wi-Fi based IoT systems. *IEEE Access* 2024.
- [11] Karras A, Giannaros A, Karras C, Theodorakopoulos L, Mammassis CS, Krimpas GA, et al. TinyML algorithms for big data management in large-scale IoT systems. *Future Internet* 2024;16(2):42.
- [12] Antonini M, Pincheira M, Vecchio M, Antonelli F. A TinyML approach to non-repudiable anomaly detection in extreme industrial environments. In 2022 IEEE International Workshop on Metrology for Industry 4.0 & IoT (MetroInd4.0&IoT) (pp. 397-402). IEEE; 2022, June.
- [13] Antonini M, Pincheira M, Vecchio M, Antonelli F. An adaptable and unsupervised TinyML anomaly detection system for extreme industrial environments. *Sensors* 2023;23(4):2344.
- [14] Akram J, Anaissi A, Othman W, Alabdulatif A, Akram A. DroneSSL: self-supervised multimodal anomaly detection in internet of drone things. *IEEE Trans Consum Electron* 2024.
- [15] Barbieri L, Brambilla M, Stefanutti M, Romano C, De Carlo N, Roveri M. A tiny transformer-based anomaly detection framework for IoT solutions. *IEEE Open Journal of Signal Processing* 2023;4:462-78.
- [16] Banerjee R, Ghose A. A lightweight deep residual network for classification of abnormal heart rhythms on tiny devices. In Joint European Conference on Machine Learning and Knowledge Discovery in Databases (pp. 317-331). Cham: Springer Nature Switzerland; 2022, September.
- [17] Ismail S, Moudoud H, Dawoud D, Reza H. Blockchain-based zero trust supply chain security integrated with deep reinforcement learning; 2024.
- [18] Hayajneh AM, Batayneh S, Alzoubi E, Alwadyan M. TinyML olive fruit variety classification by means of convolutional neural networks on IoT edge devices. *AgriEngineering* 2023;5(4):2266-83.
- [19] Singh S, Rosak-Szyrocka J, Lukács B. Design and analysis of a bandwidth aware adaptive multipath N-channel routing protocol for 5G internet of things (IoT). *Emerging Sci J* 2024;8(1):251-69.
- [20] Alajlan NN, Ibrahim DM. DDD TinyML: a TinyML-based driver drowsiness detection model using deep learning. *Sensors* 2023;23(12):5696.
- [21] Khatoon A, Wang W, Ullah A, Li L, Wang M. Optimized binary neural networks for road anomaly detection: a tinyML approach on edge devices. *Computers, Materials & Continua* 2024;80(1).
- [22] Yuan J, Elfrink S, Li Z, Cheng K, Han Q. An IoT based new platform for teaching tiny machine learning. In Proceedings of the ISCAP Conference ISSN (Vol. 2473, p. 4901); 2023.
- [23] Tharayil SM, Alrammah MM, Alghamdi MA, Aljohar FE, Otaivora WBC. Enhancing drilling equipment reliability: deep learning for predicting failure time and real-time anomaly detection. In Abu Dhabi International Petroleum Exhibition and Conference (p. D011S014R003). SPE; 2024, November.
- [24] Chiurato S. Neural network-based classification of electric vehicle acceleration pedal signals: from training to microcontroller deployment (Doctoral dissertation, Politecnico di Torino); 2023.
- [25] Le DK, Nguyen XL. Lightweight unsupervised model for anomaly detection on microcontroller platforms. *J Maritime Res* 2024;21(2):187-95.
- [26] Anidjar OH, Barak A, Ben-Moshe B, Hagai E, Tuvyah S. A stethoscope for drones: Transformers-based methods for UAVs acoustic anomaly detection. *IEEE Access* 2023;11:33336-53.
- [27] Medeiros M, Flores T, Silva M, Silva I. A multi-layered methodology for driver behavior analysis using TinyML and edge computing. In 2024 IEEE International Conference on Evolving and Adaptive Intelligent Systems (EAIS) (pp. 1-8). IEEE, 2024, May.
- [28] Jain V, Giraldo S, De Roose J, Mei L, Boons B, Verhelst M. Tinyvers: A tiny versatile system-on-chip with state-retentive eMRAM for ML inference at the extreme edge. *IEEE J Solid State Circuits* 2023;58(8):2360-71.
- [29] Razak SFA, Yogarayan S, Ullah A. Preventing impaired driving using IoT on steering wheels approach. *HighTech and Innovation Journal* 2024;5(2):400-9.
- [30] Lee KS, Kim SB, Kim HW. Enhanced anomaly detection in manufacturing processes through hybrid deep learning techniques. *IEEE Access* 2023.
- [31] Srisang W, Jaroensutasinee K, Jaroensutasinee M, Khongthong C, Piamonte JRP, Sparrow EB. PM2.5 IoT sensor calibration and implementation issues including machine learning. *Emerging Sci J*, 8(6) (2024) pp.2267-2277.
- [32] Asutkar S, Chalke C, Shivgan K, Tallur S. TinyML-enabled edge implementation of transfer learning framework for domain generalization in machine fault diagnosis. *Expert Syst Appl* 2023;213:119016.
- [33] de la Fuente R, Radrigan L, Morales AS. Enhancing predictive maintenance in mining mobile machinery through a TinyML-enabled hierarchical inference network; 2024, arXiv preprint arXiv:2411.07168.
- [34] Chen Z, Gao Y, Liang J. A self-powered sensing system with embedded TinyML for anomaly detection. In 2023 IEEE 3rd International Conference on Industrial Electronics for Sustainable Energy Systems (IESES) (pp. 1-6). IEEE; 2023, July.
- [35] Geem D, Hercules D, Pelia RS, Venkateswaran S, Griffiths A, Noe JD, et al. Progression of pediatric crohn's disease is associated with anti-tumor necrosis factor timing and body mass index Z-score normalization. *Clinical Gastroenterology and Hepatology* 2024;22(2):368-76.
- [36] Sun C, Qin W, Yun Z. A state-of-health estimation method for lithium batteries based on fennec fox optimization algorithm-mixed extreme learning machine. *Batteries* 2024;10(3):87.
- [37] Mafarja M, Thaher T, Al-Betar MA, Too J, Awadallah MA, Abu Doush I, et al. Classification framework for faulty-software using enhanced exploratory whale optimizer-based feature selection scheme and random forest ensemble learning. *Appl Intell* 2023:1-43.
- [38] Tang XF, Long YB. Integration of gradient least mean squares in bidirectional long short-term (LSTM) memory networks for metallurgical bearing ball anomaly diagnosis. *Metalurgija* 2024;63(3-4):403-6.
- [39] Mishra K, Pati J, Majhi SK. A dynamic load scheduling in IaaS cloud using binary JAYA algorithm. *J King Saud Univ-Comput Inform Sci* 2022;34(8):4914-30.
- [40] <https://research.unsw.edu.au/projects/toniot-datasets>.
- [41] Tareq I, Elbagoury BM, El-Regaily S, El-Horbaty ESM. Analysis of ton-iot, unwnb15, and edge-iiot datasets using dl in cybersecurity for iot. *Appl Sci* 2022;12(19):9572.
- [42] Saiyed MF, Al-Anbagi I. A genetic algorithm-and t-test-based system for DDoS attack detection in IoT networks. *IEEE Access* 2024;12:25623-41.
- [43] Yaras S, Dener M. IoT-based intrusion detection system using new hybrid deep learning algorithm. *Electronics* 2024;13(6):1053.
- [44] Zulfiqar Z, Malik SU, Moqurab SA, Zulfiqar Z, Yaseen U, Srivastava G. DeepDetect: An innovative hybrid deep learning framework for anomaly detection in IoT networks. *J Comput Sci* 2024;83:102426.



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